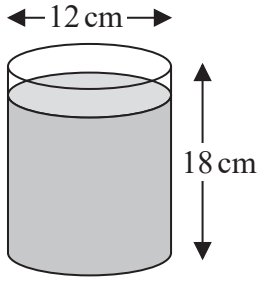
21 Joel is going to make candles.
He will pour melted wax into moulds.



Each mould is in the shape of a cylinder with diameter 12 cm and height 18 cm.

Joel has 15 kg of solid wax.
He knows that 1 kg of solid wax makes 1170 cm^3 of melted wax.

To make each candle, Joel will pour melted wax into a mould to $\frac{7}{8}$ of the height of the mould.

He wants to make as many candles as he can.

How many candles can Joel make when using 15 kg of solid wax?
You must show your working.

Radius = 12 ÷ 2 = 6 cm

Wax height = 18 ÷ 8 × 7 = 15.75 cm

1 mark
Use seven eighths of height.

Wax per candle = $\pi r^2 h = \pi \times 6^2 \times 15.75$

1 mark
Cylinder volume.

= 1781.283034... cm^3

Available wax = 15 × 1170 = 17550 cm^3

1 mark
Total melted wax.

17550 ÷ 1781.283034... = 9.852448...

1 mark
Divide total by one candle.

Only complete candles count: 9 candles.

1 mark
Round down to 9.

.....

(Total for Question 21 is 5 marks)

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